Tipping Points in Psychopathology: Using Covariance Eigenvalues as EWS for Depressive Transitions

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Problem formulation: part 1

- Symptom networks in psychopathology (Borsboom and Cramer, 2013)
  - nodes = symptoms
  - edges = how strongly symptoms interact
- Stress can enhance connections
- Transition to depressive state

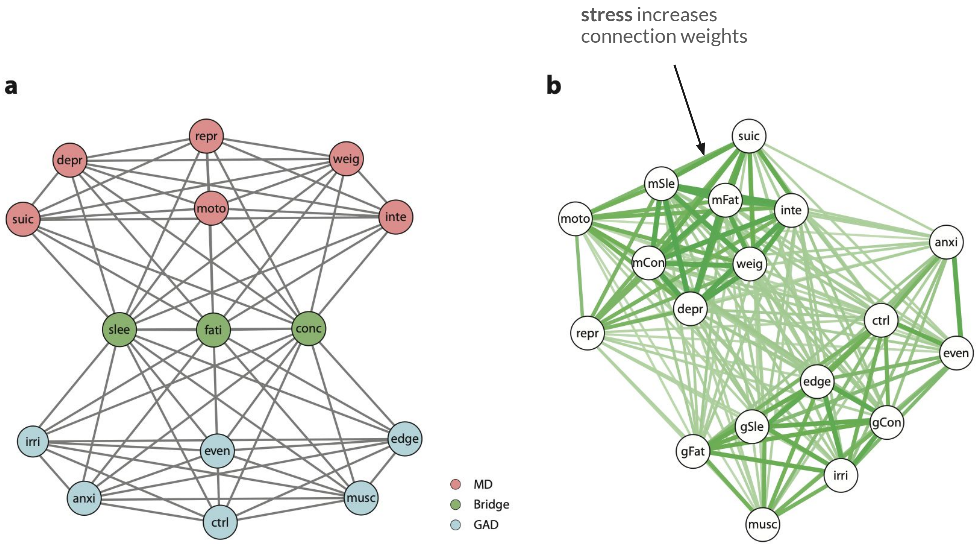


Figure 1. Networks of symptoms for MD and GAD. *From* Borsboom and Cramer (2013).

- Vulnerable vs. non-vulnerable

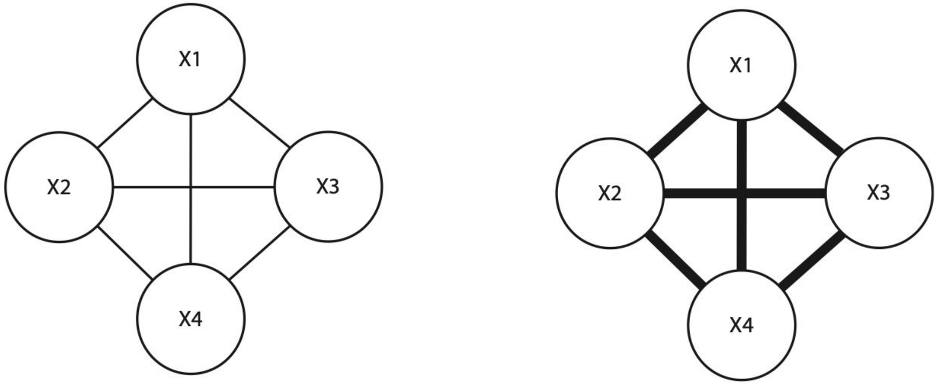
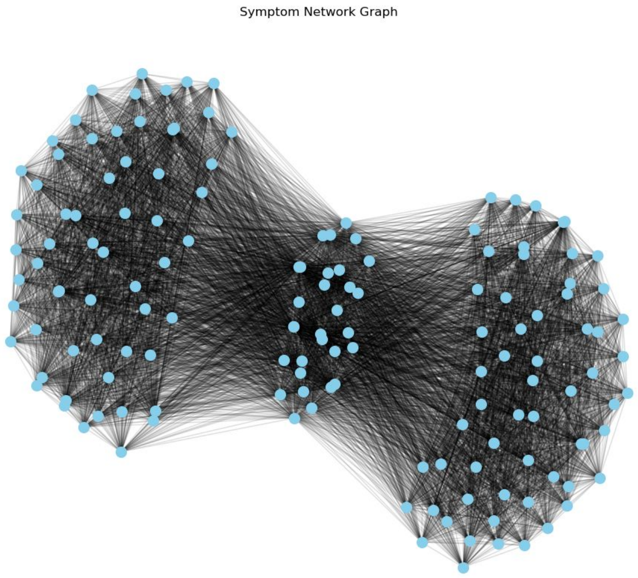


Figure 2. Weakly and strongly connected networks. *From* Borsboom and Cramer (2013).

Our network

- Borsboom & Cramer network: quickly ran into finite size effects, stochastic effects dominated
- Made it larger while maintaining cluster/bridge ratios
- 10:1 scaling



Problem formulation: part 2

- Near critical point, dominant eigenvalue of covariance matrix approaches 0 -> **C**ritical **S**lowing **D**own (CSD)
- CSD is an **E**arly **W**arning **S**ignal (EWS) -> signals transition into a new state

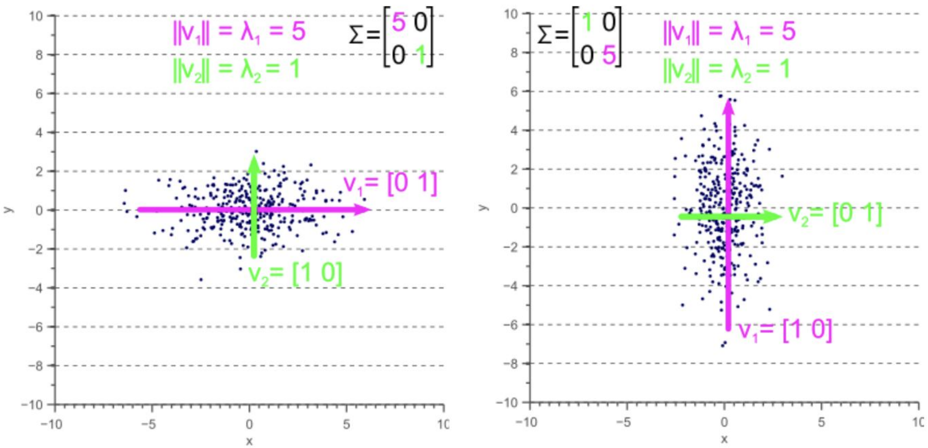


Figure 3. Eigenvalues of the covariance matrix. *From* Spruyt (2021).

Problem Intersection

Aim: study system transitions from healthy to depressive state

- on a symptom network model,
- using dominant eigenvalue of covariance matrix as the EWS.

Research Value

Research Value:

- **From Description to Prediction:** Moving beyond static checklists to dynamic complex systems
- **The Goal:** Identifying mathematical *Early Warning Signals* to predict depression before it occurs.

Key Assumptions:

- **Network Theory:** Mental disorders arise from interactions between symptoms, not just external diseases.
- **Topology**: We assume a network structure with clustered symptoms connected by a *bridge*

Limitations:

- **Synthetic Data:** This is simulated data, not real patient data.
- **Simplification:** Symptoms are modeled as binary - Active/Inactive, which goes against the continuous symptoms humans experience

Implementation

Model:

- Ising Model
- With Metropolis-Hasting to evolve the system over time.

Reasoning:

- Spins: Modeled as Binary States (+1 is Healthy, -1 is Unhealthy)
- Field (**H**): Represents Environmental Stress. Negative values represent higher stress.
- Coupling (**J**): Represents Symptom Reinforcement, how symptoms trigger each other
- Order represents overall mood/disposition, mean of node states

Network Construction:

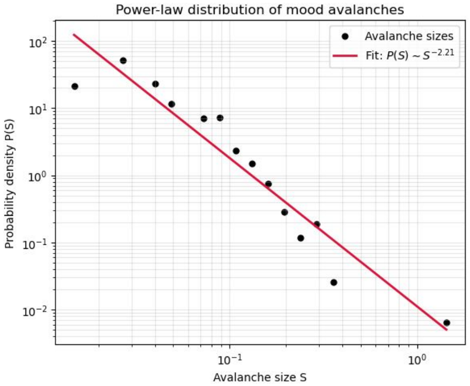
- **Topology**: Network similar to the work of *Brosboom*
- **Structure**: Two clusters of symptoms for example , Anxiety and Depression connected by Bridge Nodes

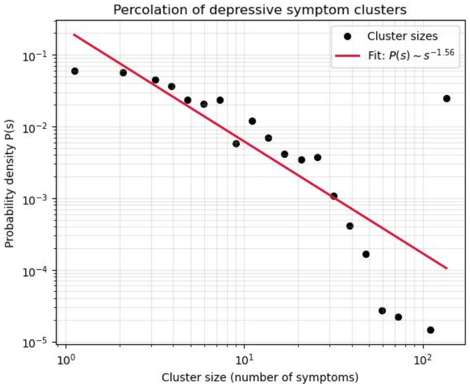
Model Validation

- **Avalanche sizes,** and
- **Percolation cluster sizes** follow a **power-law** distribution near a critical point

→

- no characteristic scale
- system close to criticality
- local interactions -> system level patterns





RQs, Hypotheses & Results

Question 1: Does the dominant eigenvalue increase as system approaches transition?

Hypotheses:

H_0: $\rho = 0 \rightarrow$ The magnitude of dominant eigenvalue does not increase with stress (no correlation)

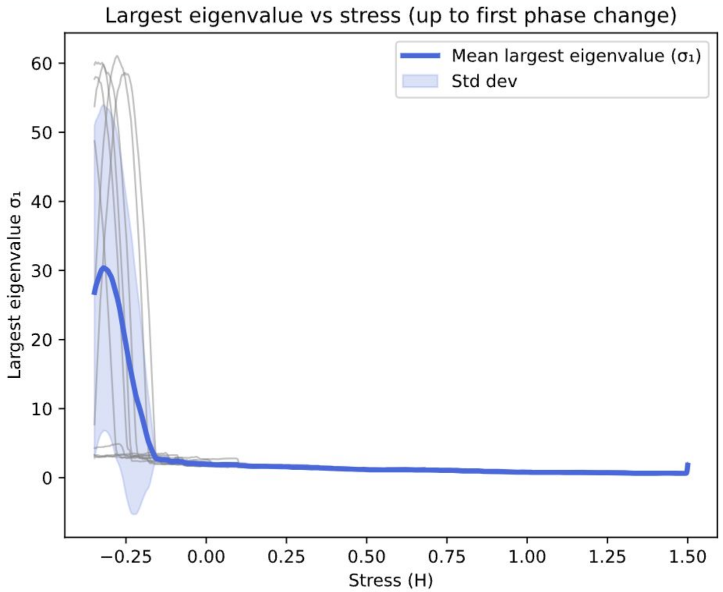
H_1: $\rho \neq 0 \rightarrow$ The magnitude of dominant eigenvalue increases with stress (there is correlation)

Question 1: Does the dominant eigenvalue increase as system approaches transition?

Test: Spearman rank correlation (ρ) between stress H and the dominant eigenvalue σ1

Result: Reject null - correlation (but not monotonic) - peaking near critical point

(p-value < 0.005)



Question 2: Does the proportion of variance explained by dominant eigenvalue increase with stress?

Hypotheses:

H_0: $\rho = 0 \rightarrow$ No correlation between stress and Proportion of Variance Explained by the dominant eigenvalue (PVE1)

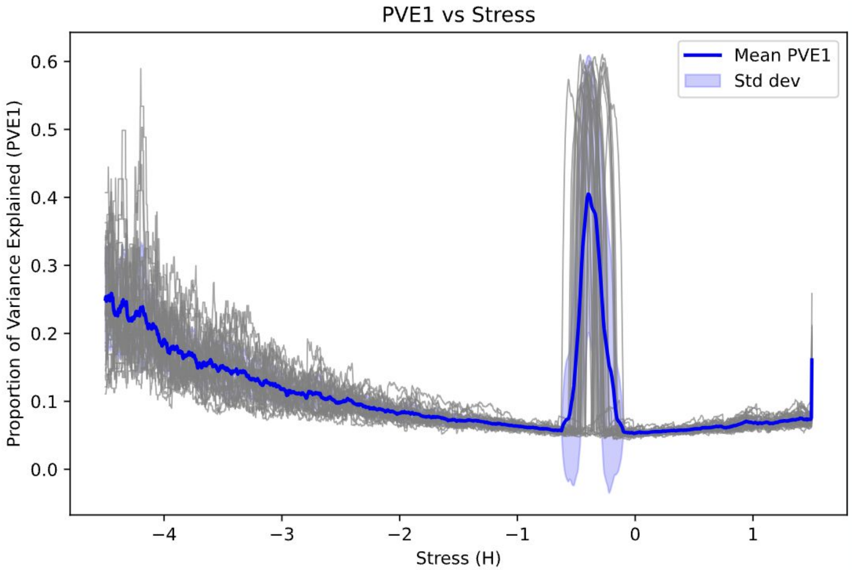
H_1: $\rho \neq 0 \rightarrow$ There is correlation between stress and PVE1

Question 2: Does the proportion of variance explained by dominant eigenvalue increase with stress?

Test: Spearman rank correlation (ρ) between stress H and PVE1

Result: Reject null

(p-value < 0.005)



Question 3: Does the dominant eigenvalue serve as an EWS for phase transition?

Hypotheses:

H_0: The dominant eigenvalue does not contain information that improves prediction of the order parameter beyond what the order's own past values provide.

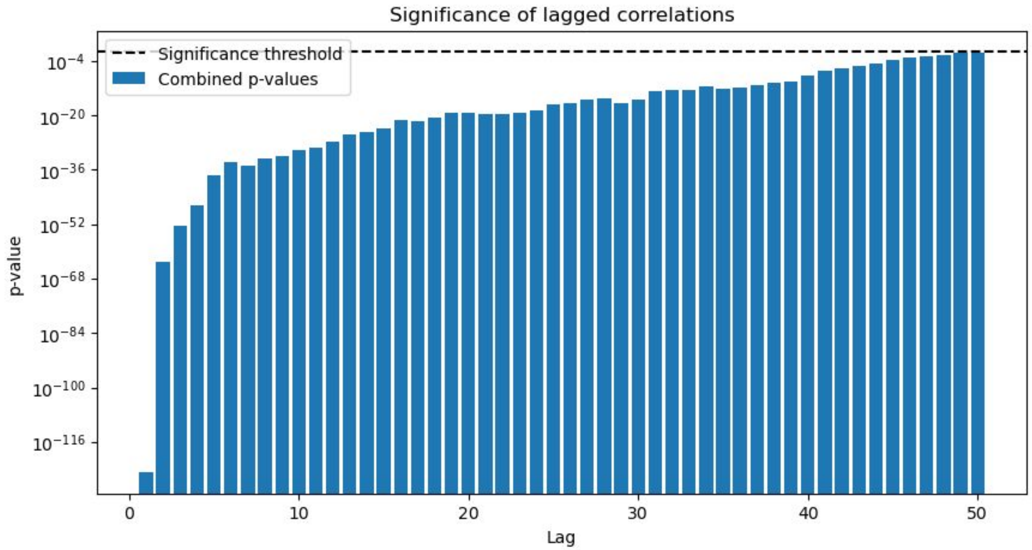
H_1: The dominant eigenvalue contains information which *does* significantly improve prediction of the order value.

Question 3: Does the dominant eigenvalue serve as an EWS for phase transition?

Results:

Granger causality test, Fisher-combined p values evaluated for 50 timesteps approaching criticality

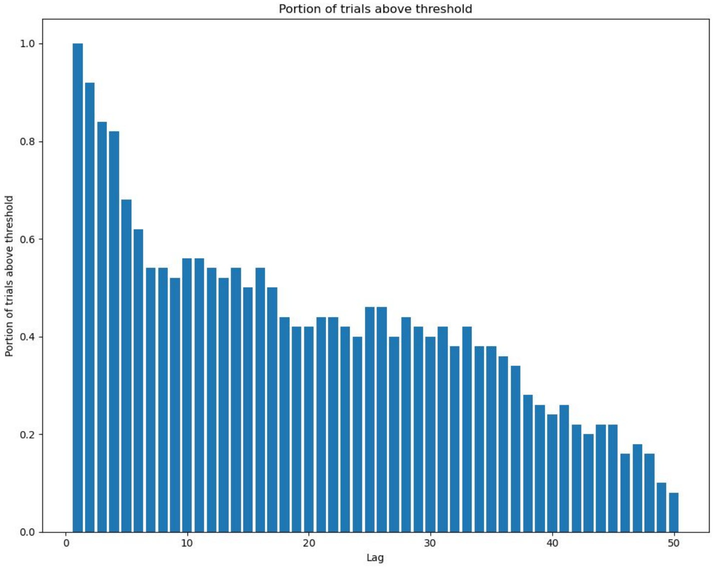
Significant results at any tested lag, **BUT…**

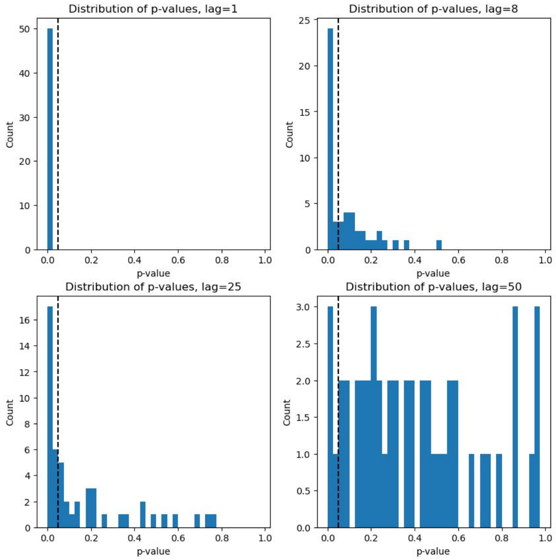


Question 3: Does the dominant eigenvalue serve as an EWS for phase transition?

Predictive power is inconsistent for higher lag values!

Question 3: Does the dominant eigenvalue serve as an EWS for phase transition?





Question 4: Does the symptom network show hysteresis?

Hypotheses:

H_0: The phase transition occurs immediately when the control parameter reaches the critical point h=0.

H_1: The phase transition exhibits hysteresis, occurring when h>0 with increasing h and h<0 with decreasing h.

Question 4: Does the symptom network show hysteresis?

Results:

One-Sample t-test:

First phase change:

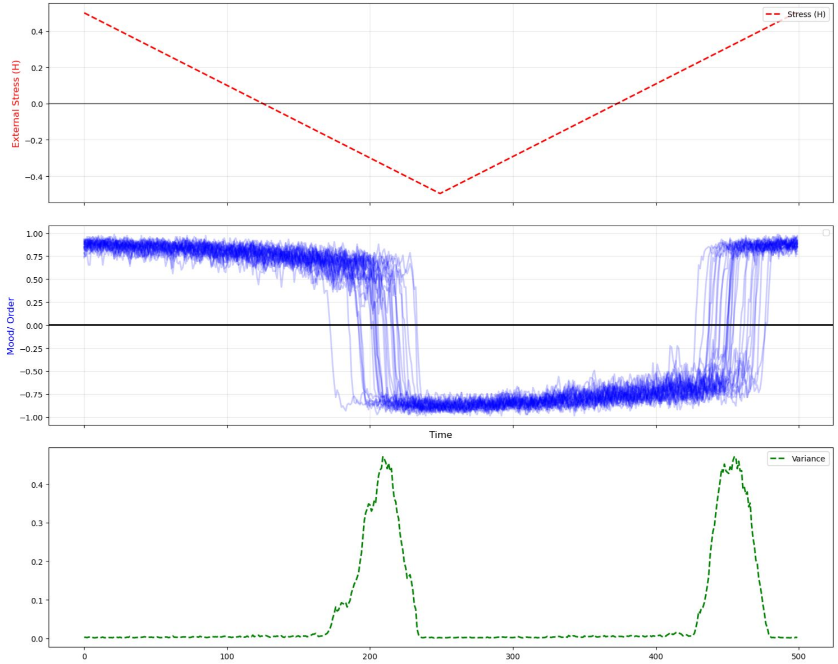
$t=26.02, p<0.005$

Second phase change:

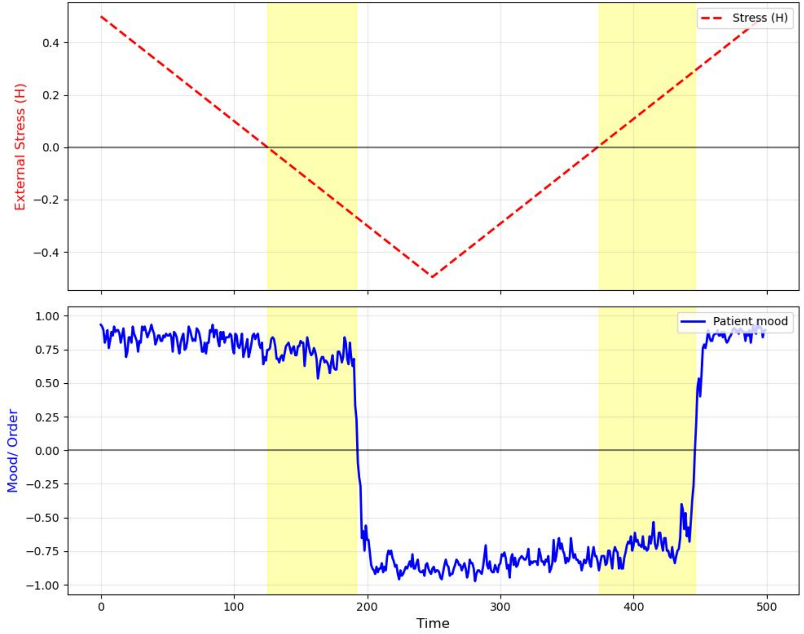
$t=15.24, p<0.005$

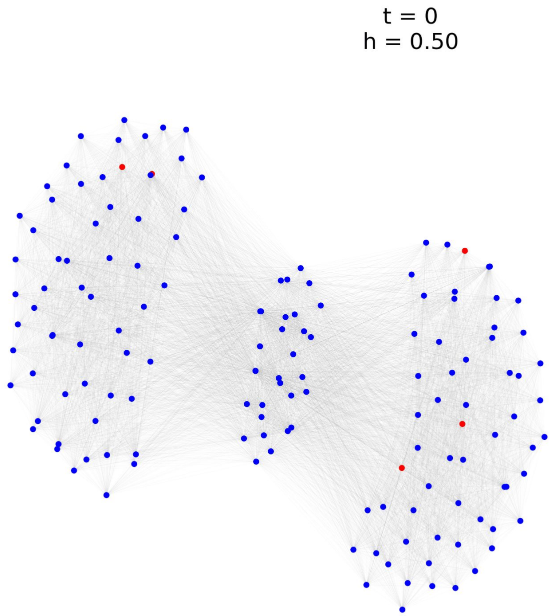
Conclusion:

Reject H_o



Question 4: Does the symptom network show hysteresis?





Main Findings

Does the dominant eigenvalue increase as system approaches transition?

- Yes, correlation peaks near the critical point.

Does the proportion of the variance explained increase with stress?

- Yes, peaks near the critical point.

Does the dominant eigenvalue serve as an EWS for phase transition?

- Yes, but less consistently as lag increases.

Does the symptom network show hysteresis?

- Yes, the network demonstrates hysteresis.

Interpretation & implications

So, what can it mean in practice?

Hysteresis -> importance of early intervention?

Future research

Personalization:

- **Calibration:** Calibrating J using data from real patients.
- **The Goal:** Moving from a generic network to a patient-specific topology.

Strategy intervention:

- **Clinical Pinning:** Simulating the effect of treating a node in the bridge, for example Insomnia to a healthy state.
- **Hypothesis:** Does stabilizing a single bridge node prevent the phase transition of the entire network?

Questions?

Thank you!

Question directions

- So, how does it all translate to mental disorders?
- Network connectivity / size
- Consistency across simulations
- Hysteresis - what does it mean in practice?

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